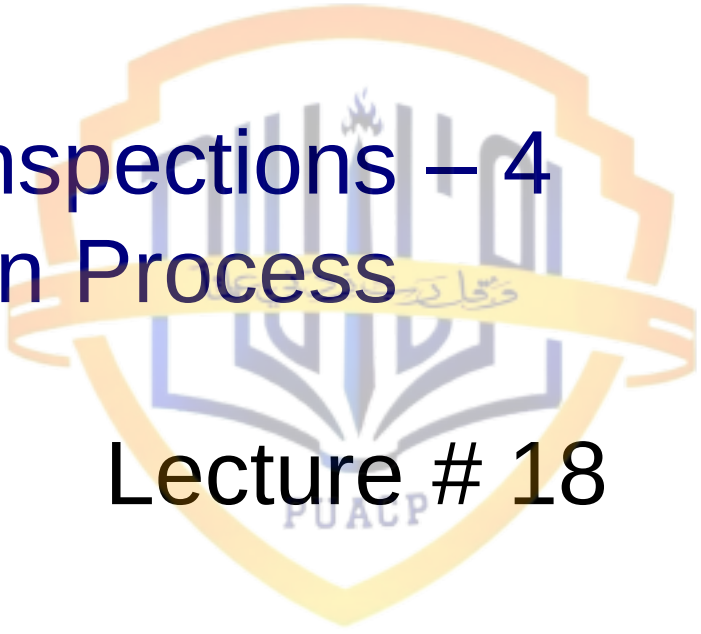




Software Inspections – 4

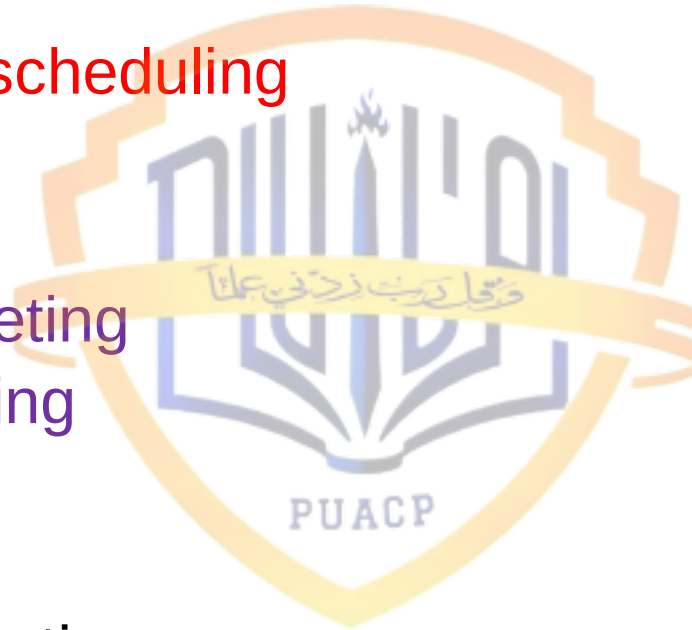
– Inspection Process



Lecture # 18

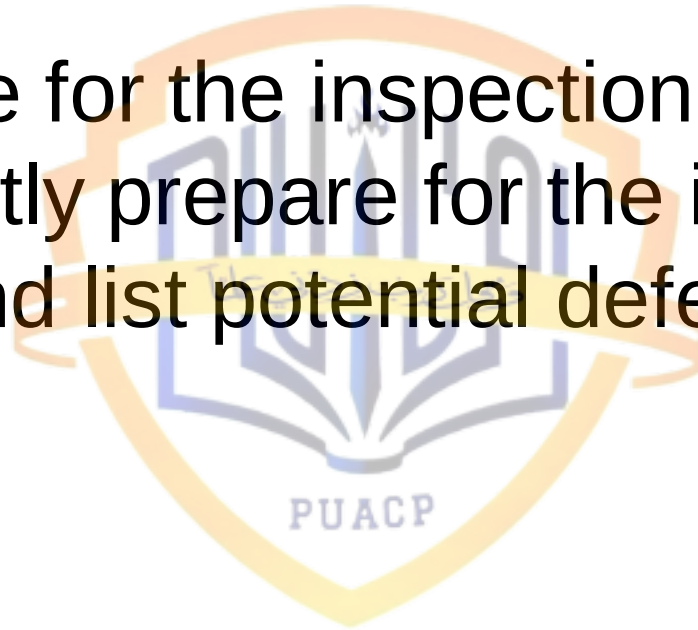
Recap and Today's Lecture: Inspection Process

- Planning and scheduling
- Overview
- Preparation
- Inspection meeting
- Analysis meeting
- Rework
- Follow-up
- Prevention meeting
- Data recording and reports
- Inspection process monitoring



Preparation - 1

- Allows time for the inspection participants to sufficiently prepare for the inspection meeting and list potential defects



Preparation - 2

- During preparation the inspectors should:
 - Increase their understanding of the material
 - Inspect the work product using the checklist appropriate to the work product
 - Identify possible defects that will be discussed at the inspection meeting
 - Create a list of minor defects and provide them to the producer
 - Note the amount of time spent in this activity

Preparation: Responsibility

- Primary responsibility is with the inspectors to ensure they have properly prepared for the inspection meeting
- If an inspector cannot prepare sufficiently, the moderator must be notified immediately and a backup inspector selected
- The inspection meeting may have to be cancelled in those situations if backup inspector is not available
- Decision should be recorded to learn during analysis

Preparation: Other Roles

- The moderator should first estimate the preparation time needed for the inspection based on the material to be inspected. These estimates should be verified with the inspection team participants
- The moderator needs to get a commitment from each participant that enough time is allocated and that it will be sufficient for him/her to prepare

Preparation: Entry Criteria - 1

- The overview, if needed, has been satisfactorily completed
- Any open issues identified for the overview have been closed and addressed in the work product or are documented as open issues and provided as ancillary material for the preparation

Preparation: Entry Criteria - 2

- Open issues not closed are documented for tracking within the change control system used by the project
- The producer determines that the work product is ready for inspection
- The work product has reached closure and the code complies with defined standards, style guides, and templates for format

Preparation: Entry Criteria - 3

- All necessary ancillary material have been made available well in advance
- The work product includes all base-lined function and approved changes for this planned work product completion date
- The amount of time needed for preparation has been confirmed with the inspectors and is available to them

Preparation: Entry Criteria - 4

- Predecessor and dependent work products are available, have been inspected, and meet exit criteria
- The moderator and producer have defined the coverage of material to be inspected
- The work products allow easy identification of defects by location in the material
- The moderator agrees that the work product is inspectable

Preparation: Tasks - 1

- Each inspector uses the scheduled time to complete the preparation in a style and format they are comfortable with
- The material to be inspected is marked with questions, concerns, and possible defects, both major and minor, found during inspection

Preparation: Tasks - 2

- The minor defects are either recorded on a separate sheet that will be delivered to the moderator at the start of the inspection meeting or they are clearly noted in the marked material that will be delivered to the moderator at the end of the inspection meeting. Each minor defect should be noted by location in the work product when using a minor list

Preparation: Validation/Verification - 1

- The moderator uses the preparation entry criteria and procedure
- The moderator uses the minor defect information to determine if all inspectors have properly performed preparation
- The inspectors have confirmed that they have prepared

Preparation: Validation/Verification - 2

- The SQA group ensures that the moderator has used the preparation procedure and that the inspectors performed sufficient preparation. This can be done via audits of the process records or sampling of inspections

Preparation: Validation/Verification - 3

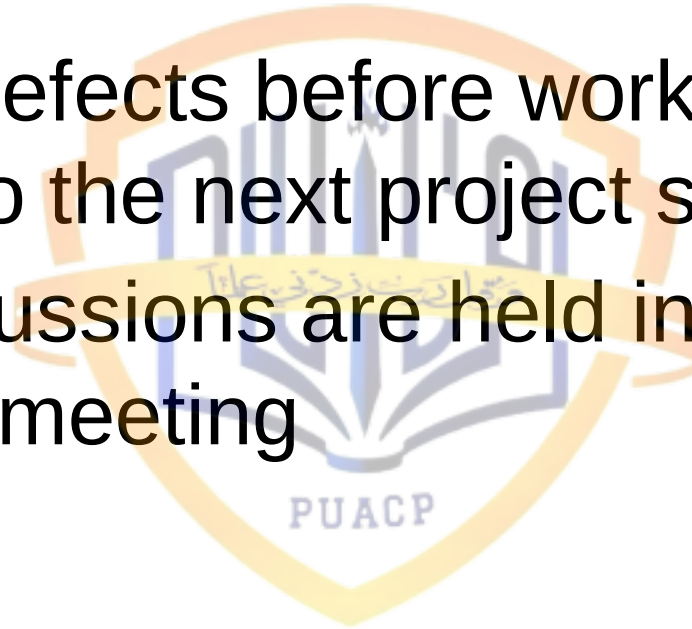
- Data gathered in this activity
 - How much time was spent in preparation
 - How long a period between notification of the inspection and the preparation
 - How many inspection meetings required rescheduling due to insufficient preparation
 - The number of major and minor defects found during preparation

Preparation: Exit Criteria

- Each inspector has completed sufficient preparation based on organization and project preparation time criteria
- Minor defect inputs are complete
- Preparation notes are recorded on the work product materials or defect lists

Inspection Meeting - 1

- Identifies defects before work product is passed into the next project stage
- Some discussions are held in the inspection meeting

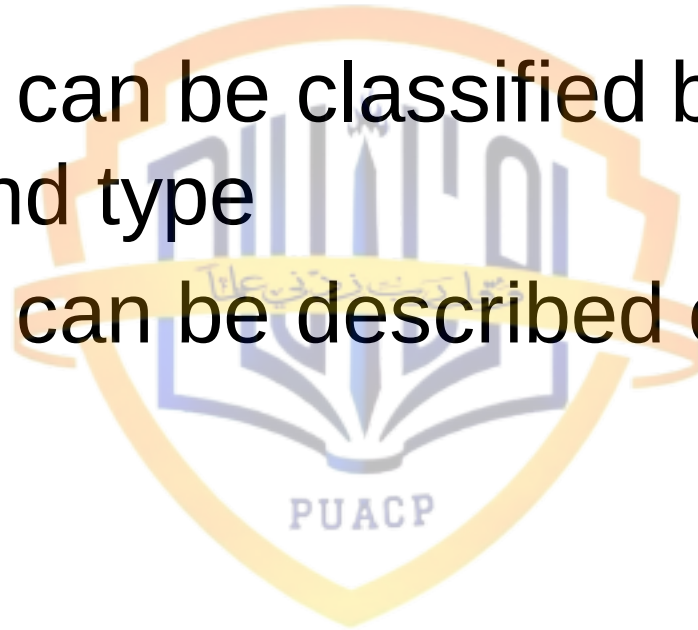


Inspection Meeting - 2

- The identified defect is agreed to be a defect, or at least a potential defect, by the inspection team, including the producer
- If a discussion item cannot be agreed to be a defect, it should be noted as an open issue to be resolved after the meeting

Inspection Meeting - 3

- The defect can be classified by class, severity, and type
- The defect can be described crisply but sufficiently

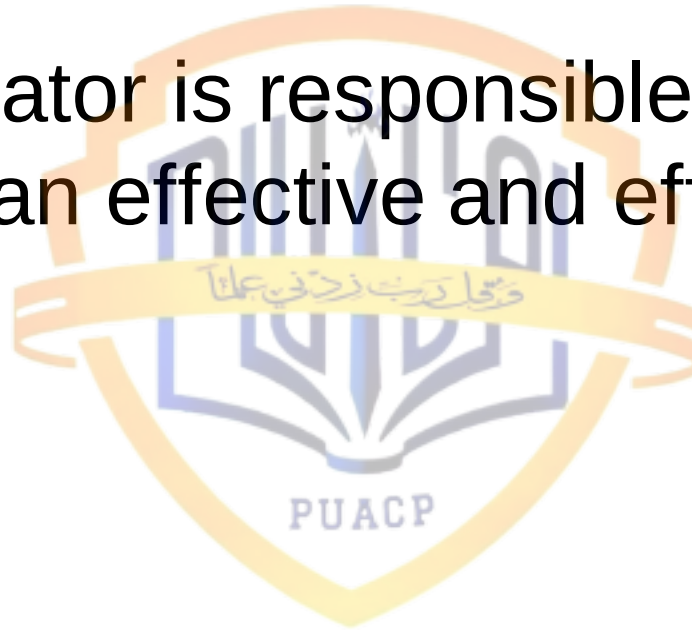


Inspection Meeting - 4

- The inspection meeting has schedule and entry requirements. If inspectors are late by ten minutes, postponement should be considered
- The critical inspectors include the moderator, producer, and reader

Inspection Meeting: Responsibility

- The moderator is responsible for managing an effective and efficient meeting



Inspection Meeting: Other Roles - 1

- The producer is responsible for the inspected work product, answering questions, and concurring on identified defects or adequately explaining to the inspection team's agreement why the identified possible defect is not a defect

Inspection Meeting: Other Roles - 2

- The reader is responsible for focusing and pacing the inspection meeting by leading the team through the material
- The recorder is responsible for correctly recording all identified defects and open issues
- All inspectors, including the producer, are responsible for sharing their questions and identified defects found during preparation, and work together to find more defects in meeting

Inspection Meeting: Entry Criteria - 1

- The inspection team members are sufficiently present in number and role assignments
- Inspection materials were available for preparation with sufficient time for study and review before the inspection meeting, including necessary reference material

Inspection Meeting: Entry Criteria - 2

- Inspectors have adequately prepared
- Inspectors have submitted their minor defects list at the start of the meeting or have marked the work products that will be provided at the end of the meeting
- Scope of the inspection meeting has been defined

Inspection Meeting: Entry Criteria - 3

- Recorder and a data recording system are available
- Other roles; e.g., reader have been assigned
- The producer has identified any new potential problem areas

Inspection Meeting: Tasks - 1

- Brief introduction (moderator)
- Preparedness check (moderator)
- Read the work product (reader)
- Identify defects (inspectors)
- Record defects (recorder)

Inspection Meeting: Tasks - 2

- Determine disposition of material (inspection team)
 - Accept the material
 - Accept the material after verification with follow-up inspector
 - Request the work product to be re-inspected after rework
 - Recommend re-engineering of the work product followed by a new inspection

Inspection Meeting: Validation/Verification - 1

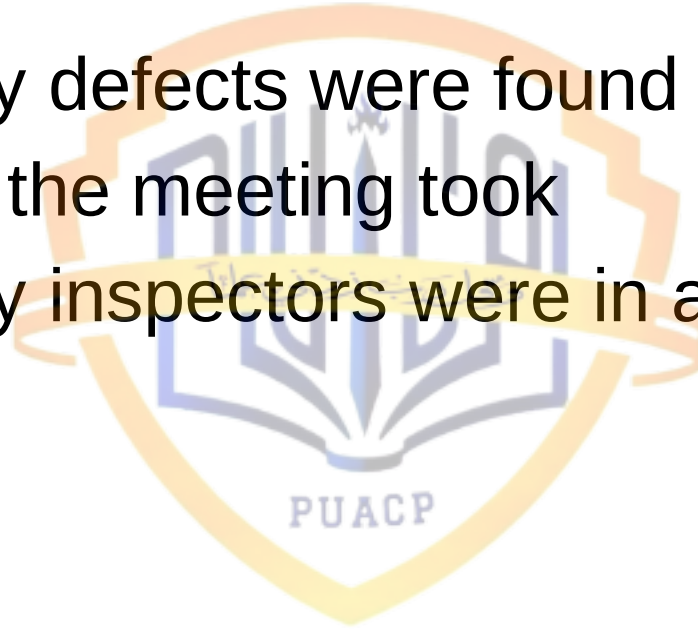
- The moderator, using the inspection meeting entry criteria and procedure, determines if the team has properly performed the inspection
- The inspectors participated in an effective meeting
- The SQA group ensures that inspection meeting procedure and that the inspectors performed sufficient preparation. This can be done via audits of the process records or sampling of inspections

Inspection Meeting: Validation/Verification - 2

- Data gathered during this activity
 - How much time was spent in the inspection meeting
 - How long a period between the preparation and the inspection meeting
 - How many inspection meetings required rescheduling due to insufficient preparation
 - How many inspections required re-inspection

Inspection Meeting: Validation/Verification - 3

- ☐ How many defects were found
- ☐ How long the meeting took
- ☐ How many inspectors were in attendance



Inspection Meeting: Exit Criteria - 1

- The inspection materials have been inspected and coverage of the work product is completed as planned
- The inspection results fall within expected tolerance of performance for
 - Time spent during preparation
 - Time spent at the inspection meeting
 - Defect density

Inspection Meeting: Exit Criteria - 2

- The defects and the conduct of the inspection have been recorded and the team concurs with the contents
- Open issues have been recorded for follow-up during rework
- The moderator or a designee has been appointed to perform follow-up with the producer

Inspection Meeting: Exit Criteria - 3

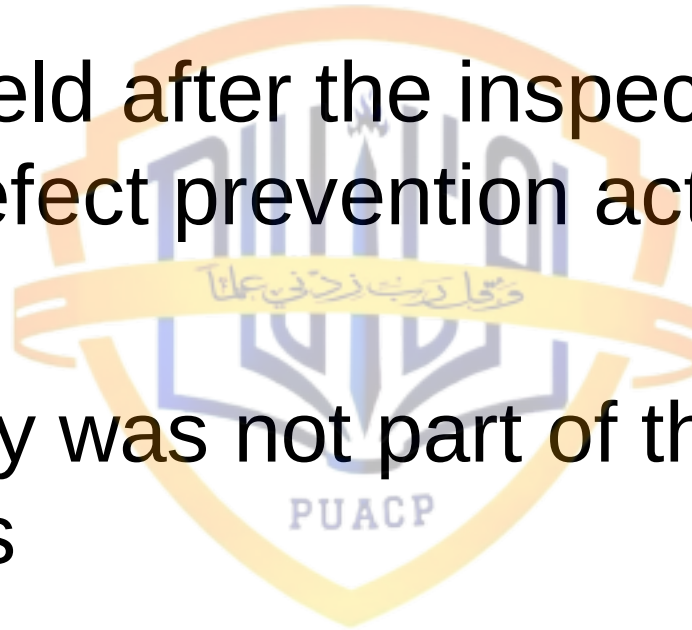
- Data is available to update the process data base
- Any associated deviations or risks are noted
- Decisions to re-inspect or not have been reviewed against criteria
- Decision on re-engineering has been addressed

Inspection Meeting: Exit Criteria - 4

- Process defects have been recorded, as appropriate, as well as product defects
- The locations of the defects of the inspected work product are clearly noted to facilitate repair
- A decision is taken on the timeframe by which defect repairs and open issues will be resolved
- The inspection satisfies the criteria to be indicated as performed

Analysis Meeting

- Which is held after the inspection meeting, to begin defect prevention activities
- This activity was not part of the original inspections



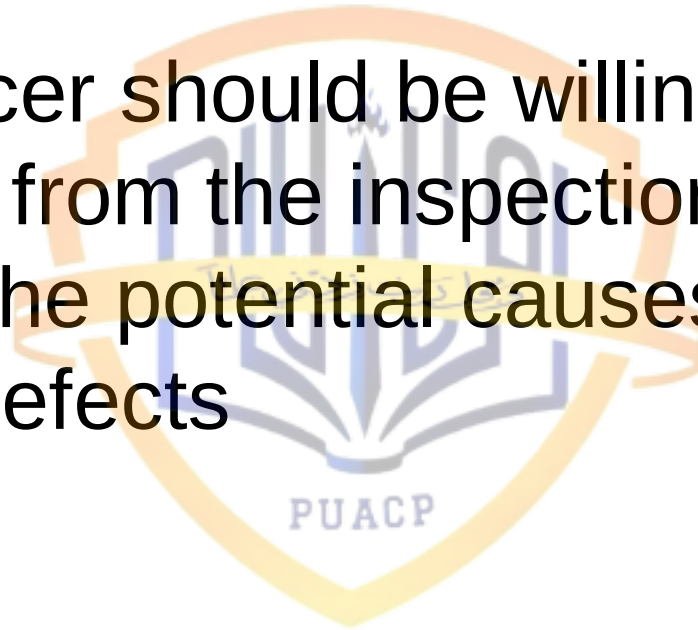
Analysis Meeting: Responsibility

- The moderator with the project lead determines whether this activity will be performed



Analysis Meeting: Other Roles

- The producer should be willing to accept open input from the inspection team regarding the potential causes of the identified defects



Analysis Meeting: Entry Criteria

- The project lead and moderator have chosen this activity to be performed
- The inspection team has been trained in causal analysis techniques. Training in team dynamics and group behavior can be helpful
- Major defects have been found during the inspection
- A defect taxonomy or set of cause categories has been defined

Analysis Meeting: Tasks

- Select the defects to discuss
- Determine the potential causes of the defects discussed
- The recorder will record the analysis meeting results and provide them to the inspection coordinator or SEPG as input for process improvement consideration within the organization at the prevention meeting

Analysis Meeting: Validation/Verification - 1

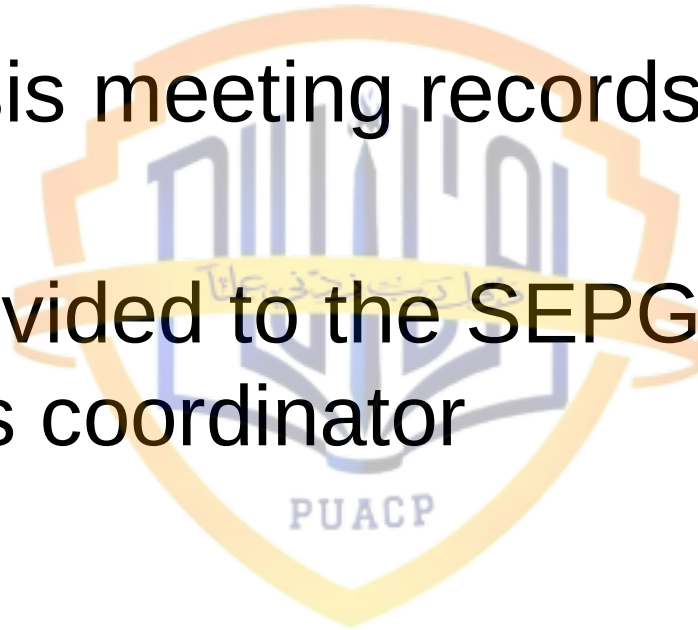
- The moderator uses the analysis meeting entry criteria and procedure to determine if all inspectors have properly participated and the meeting was effective
- The inspectors have participated
 - If they cannot participate, they must notify the moderator at the start of the inspection meeting
- The SQA group ensures that the moderator has used the Analysis meeting checklist and reviews the recorder's report for sufficiency. Audits

Analysis Meeting: Validation/Verification - 2

- Data gathered during this activity
 - How much time was spent in the analysis meeting
 - How many defects were discussed
 - How many defects were assigned causes

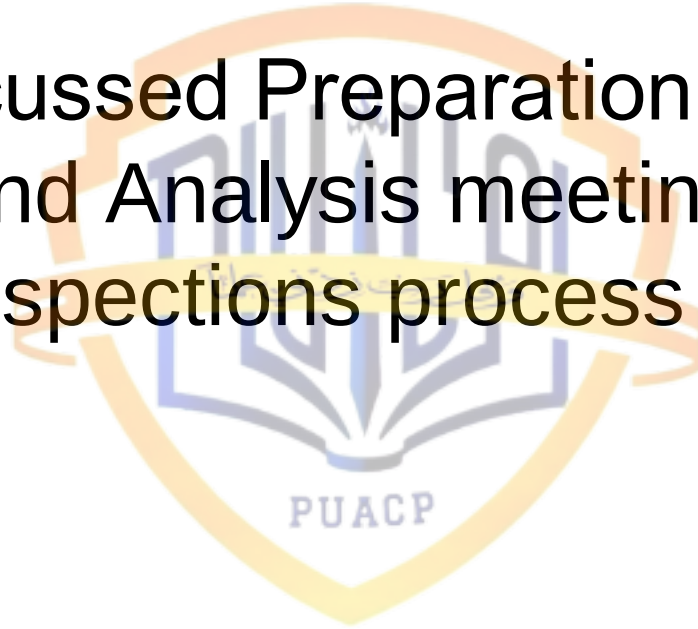
Analysis Meeting: Exit Criteria

- The analysis meeting records have been completed
- Data is provided to the SEPG or inspections coordinator



Summary

- We've discussed Preparation, Inspection meeting, and Analysis meeting of the software inspections process in this lecture



References

- Software Inspections by Ronald A. Radice, Tata-McGraw Hill, 2003 (Chapter 3.5-3.7)

